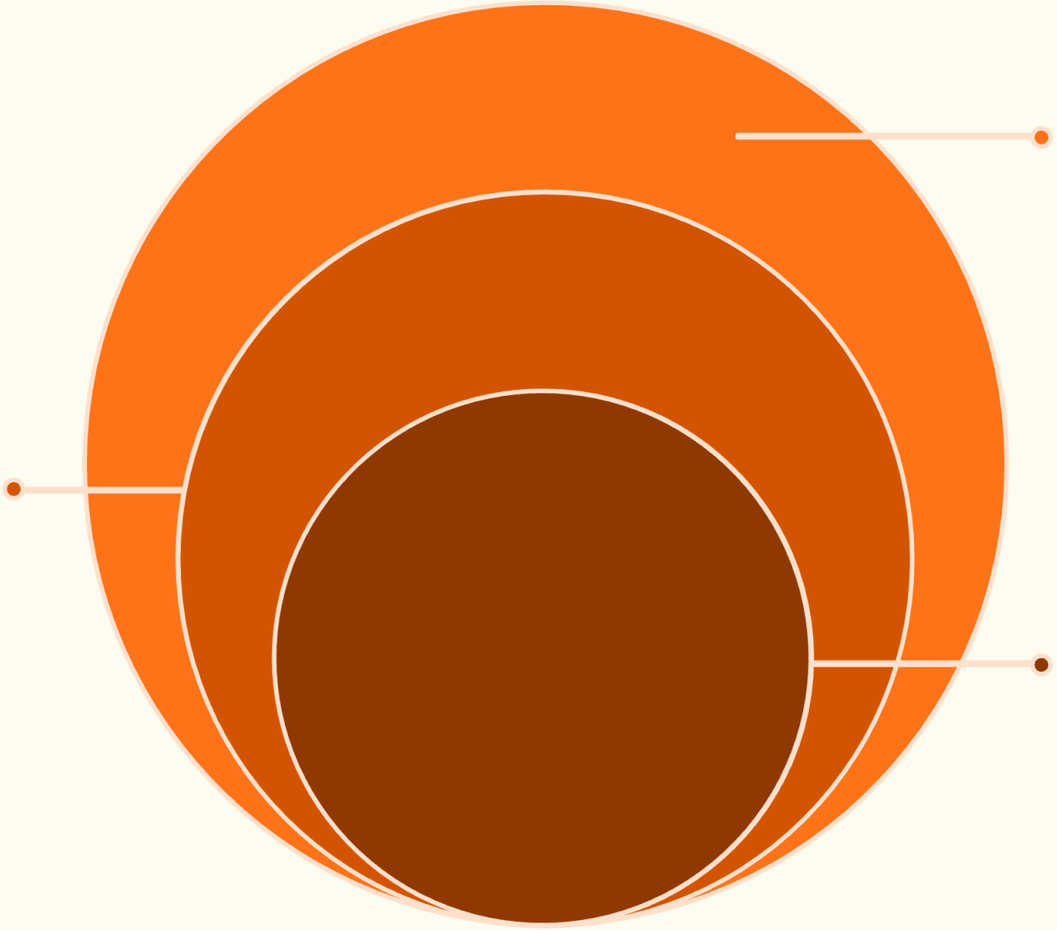
Vehicle Rental Management System

Problem: Manual rental management leads to booking errors, poor tracking, and incorrect cost calculations.

Solution: A desktop application that automates booking, tracks vehicle availability, and calculates rental costs accurately.

Target Users: Small vehicle rental shops, administrators,tourists

System Architecture



GUI Layer(Main Frame,Vehicle rental app)

User interaction and error display.

Service Layer (rentalservice)

Handles booking logic and coordination.

Model Layer (vehicle, customer, booking)

Stores data and business entities

VEHICLE RENTAL APP

Object-Oriented Programming Concepts

Abstraction

Vehicle is an abstract class used to define common properties and methods for all vehicle types.

Inheritance

Car, Bike, and Truck extend Vehicle to reuse common attributes and behavior.

Polymorphism

Each vehicle type overrides calculateRentalCost() to implement different pricing logic.

Encapsulation

Private fields with public getters/setters are used in model classes like vehicle and customer.

Collections

HashMap and ArrayList are used in rentalservice to store vehicles, customers, and bookings.

Exception Handling

Custom exceptions handle invalid booking duration and unavailable vehicles during booking.

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Technical Highlight

Booking Logic

Service layer checks vehicle availability before creating bookings.

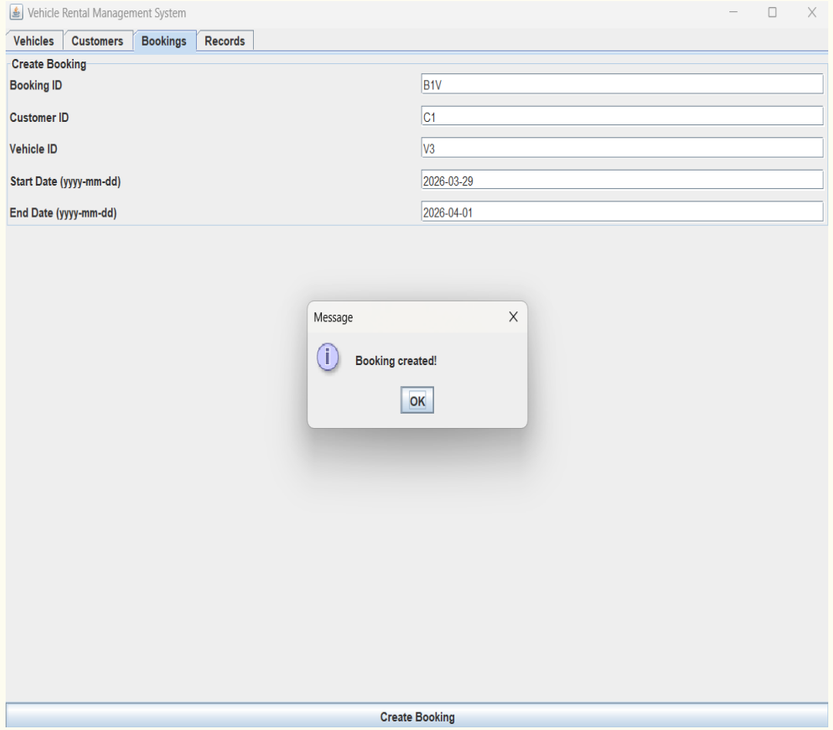
Why This Matters

Prevents double bookings, maintains data consistency, and ensures realistic business operations.

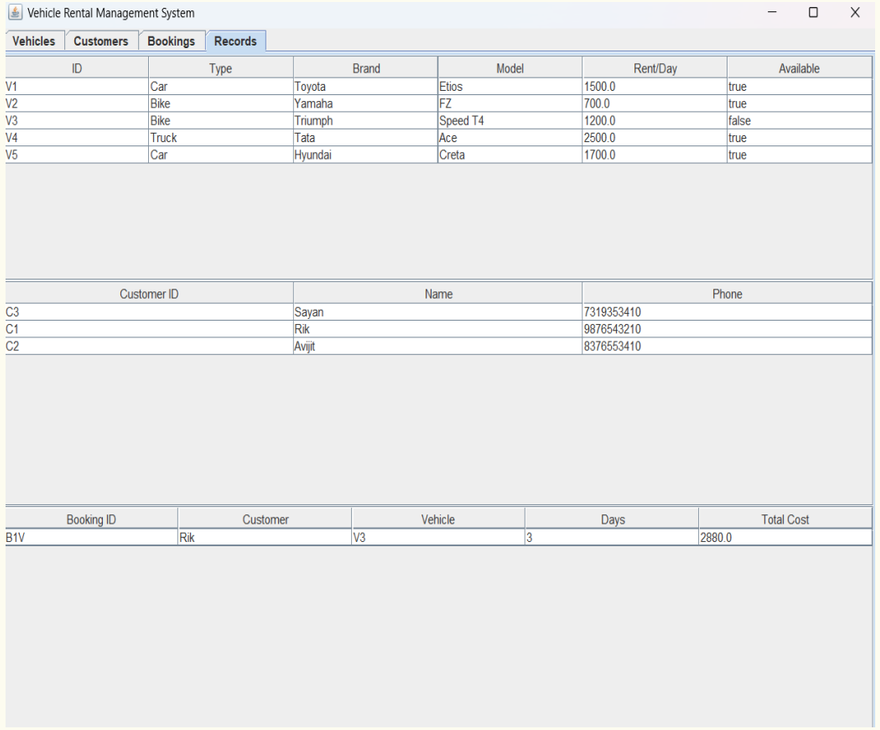
Design Decision

Business logic is separated from the GUI using a service layer, improving modularity, scalability, and maintainability.

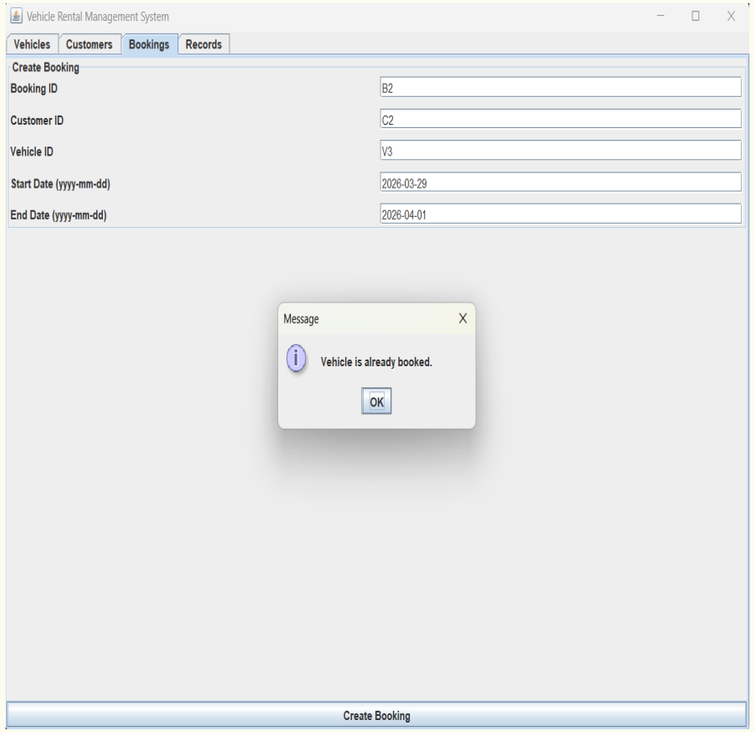
Live Demo



Initializing a booking



Vehicels,Customer & Booking records



Custom exception

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